Durgesh Rajpurohit

B.Tech CSE (Artificial Intelligence) | BS Data Science, IIT Madras

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PROFESSIONAL SUMMARY

Computer Science student specializing in Artificial Intelligence, pursuing a parallel BS in Data Science at IIT Madras. Interested in applied AI, geospatial and satellite-imagery analysis, and software development. Hands-on project work spans AI-driven reasoning, maritime intelligence and land-cover recognition, alongside a published paper on mitigating algorithmic bias and ongoing research on rare earth elements and SAR/Sentinel imagery.

EDUCATION

Guru Gobind Singh Indraprastha University (GGSIPU), Delhi *Currently pursuing*
B.Tech in Computer Science and Engineering, specialization in Artificial Intelligence

Indian Institute of Technology Madras *Currently pursuing*
BS Degree Programme in Data Science

TECHNICAL SKILLS

Programming & Development	C, C++, JavaScript, React, Vite, CSS, Frontend Development
AI, Data & Research Areas	Artificial Intelligence, Machine Learning, Data Science, NLP, OCR, Image Analysis
Geospatial & Satellite	Geospatial Data, GIS, Remote Sensing, Satellite Imagery Analysis, Vessel-Data Correlation
Additional	Poster & Visual Design, Team Collaboration, Technical Communication

SELECTED PROJECTS

MultiMind AI | Streamlit application *Hacknovate Finalist*
- Developed an AI-powered application for multi-perspective reasoning and structured, debate-style discussion.
- Reached the finalist stage at Hacknovate.

Satellite-Based Maritime Intelligence *Ongoing*
- Exploring satellite imagery, geospatial data and vessel information for oil-spill detection and maritime monitoring, linking AI methods with remote sensing and vessel-data correlation.

Git Bhu: Geospatial Land-Record Management *Ongoing*
- Exploring how land-parcel information, historical land records and traceable changes can be organized in a geospatial information-management system.

AI-Based Land-Cover Recognition
- Applied AI-based techniques to recognize and classify land-cover categories from imagery.

Career Website | React, Vite, CSS, Framer Motion *In development*
- Building a frontend site that presents career information and early-career resources to help users explore job opportunities.

Technical Explorations *Exploratory*
- eRTMAC-NWIS: historical drilling-report analysis with NLP/OCR, nearby-well intelligence, geospatial mapping and predictive drilling-risk concepts.
- Quantum-inspired route optimization: vehicle-routing and shortest-path problems under varying traffic conditions.

RESEARCH & PUBLICATIONS

AI Bias and Fairness *Published, 2026*
- Published paper: “Mitigating Algorithmic Bias through Disparate Impact Ratio: A Pre-processing Approach to Fair Machine Learning.”
- Authors: Durgesh Rajpurohit, Vanshika Thakur, Dhruv Upreti, Nikhil, Aarzu. ICAMC 2026, HMRITM.

Rare Earth Elements *Ongoing*
- Conducting ongoing research related to rare earth elements.

SAR and Sentinel Satellite Imagery *Ongoing*
- Exploring research that uses SAR and Sentinel satellite imagery; work in progress.

PROFESSIONAL EXPERIENCE

Website Development & Data Management | NRC Global *Ongoing*
- Working on website development and data management for NRC Global.

LEADERSHIP & ACTIVITIES

- **Class Representative** (~1 year, ongoing): Communicate student concerns and help coordinate academic information between students and faculty.
- **Core Member, Web Team** (HMRITM College Fest): Part of the web team for the college fest.
- **Event Design**: Created posters and visual materials for college events.
- **Educational Industrial Visit** (Yakult factory, Sonipat): Helped organize an educational visit for students.